

Long-term dietary sodium, potassium and fluid intake; exploring potential novel risk factors for renal cell cancer in the Netherlands Cohort Study on diet and cancer

Background: As sodium, potassium and fluid intake are related to hypertension, an established risk factor for renal cell cancer (RCC), they may be independent risk factors for RCC. Methods: The Netherlands Cohort Study (NLCS) with case-cohort design included 120 852 participants aged 55–69 years. At baseline, diet and lifestyle were assessed with questionnaires. After 17.3 years of follow-up, 485 RCC cases and 4438 subcohort members were available for analyses. Results: Sodium intake increased RCC risk (P -trend=0.03), whereas fluid and potassium intake did not. For high sodium and low fluid intake, the RCC risk additionally increased (P -interaction=0.02). Conclusion: Sodium intake is a potential risk factor for RCC, particularly if fluid consumption is low.